

Exercise 3.1

1. Write the following sets in set builder notation:

(i) $\{1, 4, 9, 16, 25, 36, \dots, 484\}$

$$\{1, 4, 9, 16, 25, 36, \dots, 484\}$$

$$= \{x | x = n^2, n \in N \wedge 1 \leq n \leq 22\}$$

Note: This set consists of perfect squares from 1^2 to 22^2 (since $22^2 = 484$).

(ii) $\{2, 4, 8, 16, \dots, 256\}$

$$\{2, 4, 8, 16, \dots, 256\}$$

$$= \{x | x = 2^n, n \in N \wedge 1 \leq n \leq 8\}$$

Note: This set consists of powers of 2 from 2^1 to 2^8 (since $2^8 = 256$).

(iii) $\{0, \pm 1, \pm 2, \dots, \pm 1000\}$

$$\{0, \pm 1, \pm 2, \dots, \pm 1000\}$$

$$= \{x | x \in Z \wedge -1000 \leq x \leq 1000\}$$

Note: This set includes all integers from -1000 to 1000 .

(iv) $\{6, 12, 18, \dots, 120\}$

$$\{6, 12, 18, \dots, 120\}$$

$$= \{x | x = 6n, n \in N \wedge 1 \leq n \leq 20\}$$

Note: This set consists of multiples of 6 from 6×1 to 6×20 (since $6 \times 20 = 120$).

(v) $\{100, 102, 104, \dots, 400\}$

$$\{100, 102, 104, \dots, 400\}$$

$$= \{x | x = 100 + 2n, n \in W \wedge 0 \leq n \leq 150\}$$

Note: This set includes even numbers from 100 to 400.

(vi) $\{1, 3, 9, 27, 81, \dots\}$

$$\{1, 3, 9, 27, 81, \dots\}$$

$$= \{x | x = 3^n, n \in W\}$$

Note: This set consists of powers of 3 starting from $3^0 = 1$.

(vii) $\{1, 2, 4, 5, 10, 20, 25, 50, 100\}$

$$\{1, 2, 4, 5, 10, 20, 25, 50, 100\}$$

$$= \{x | x \text{ is divisor of } 100, n \in N \wedge 1 \leq x \leq 100\}$$

Note: This set includes all positive divisors of 100.

(viii) $\{5, 10, 15, \dots, 100\}$

$$\{5, 10, 15, \dots, 100\}$$

$$= \{x | x = 5n, n \in N \wedge 1 \leq n \leq 20\}$$

Note: This set consists of multiples of 5 from 5×1 to 5×20 (since $5 \times 20 = 100$).

(ix) The set of all integers between -100 and 1000 .

$$= \{x | x \in Z \wedge -100 \leq x \leq 1000\}$$

Note: This set includes all integers from -100 to 1000 .

2. Write each of the following sets in tabular form:

(i) $\{x | x \text{ is a multiple of } 3 \wedge x \leq 36\}$

$$\{x | x \text{ is a multiple of } 3 \wedge x \leq 36\}$$

$$= \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36\}$$

(ii) $\{x | x \in R \wedge 2x + 1 = 0\}$

$$2x + 1 = 0$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

$$\{x | x \in R \wedge 2x + 1 = 0\}$$

$$= \left\{-\frac{1}{2}\right\}$$

(iii) $\{x | x \in P \wedge x < 12\}$

$$\{x | x \in P \wedge x < 12\}$$

$$= \{2, 3, 5, 7, 11\}$$

(iv) $\{x | x \text{ is a divisor of } 128\}$

$$\{x | x \text{ is a divisor of } 128\}$$

$$= \{1, 2, 4, 8, 16, 32, 64, 128\}$$

(v) $\{x | x = 2^n, n \in N \wedge n < 8\}$

$$\{x | x = 2^n, n \in N \wedge n < 8\}$$

$$= \{2, 4, 8, 16, 32, 64, 128\}$$

(vi) $\{x | x \in N \wedge x + 4 = 0\}$

$$\{x | x \in N \wedge x + 4 = 0\}$$

$$= \{ \}$$

As $x + 4 = 0 \Rightarrow x = -4 \notin N$

(vii) $\{x | x \in N \wedge x = x\}$

$$\{x | x \in N \wedge x = x\}$$

$$= \{1, 2, 3, \dots\}$$

(viii) $\{x | x \in Z \wedge 3x + 1 = 0\}$

$$\{x | x \in Z \wedge 3x + 1 = 0\}$$

$$= \{ \}$$

As $3x + 1 = 0 \Rightarrow x = -\frac{1}{3} \notin Z$

3. Write two proper subsets of each of the following sets:

(i) $\{a, b, c\}$

The Proper subsets of $\{a, b, c\}$ are $\{a\}, \{b\}$

(ii) $\{0, 1\}$

The Proper subsets of $\{0, 1\}$ are $\{0\}, \{1\}$

(iii) N

The Proper subsets of N are Prime Numbers, Even Natural Numbers

(iv) Z

The Proper subsets of Z are Prime Numbers, Whole Numbers

(v) Q

The Proper subsets of Q are Integers, Natural Numbers

(vi) $\{x|x \in Q \wedge 0 < x \leq 2\}$

The Proper subsets of $\{x|x \in Q \wedge 0 < x \leq 2\}$ are $\{\frac{1}{2}\}$, $\{\frac{3}{4}\}$

4. Is there any set which has no proper subset? If so, name that set.

Yes, there exist a set which has no proper subset, that is called 'empty set' $\{\}$.

5. What is the difference between $\{a, b\}$ and $\{\{a, b\}\}$?

$\{a, b\}$ set has two elements: a and b . While $\{\{a, b\}\}$ set has only one element $\{a, b\}$.

6. What is the number of elements of the power set of each of the following sets?**(i) $\{\}$**

$$\begin{aligned}\text{No. of elements of the power set} &= 2^n \\ &= 2^0 \\ &= 1\end{aligned}$$

(ii) $\{0, 1\}$

$$\begin{aligned}\text{No. of elements of the power set} &= 2^n \\ &= 2^2 \\ &= 4\end{aligned}$$

(iii) $\{1, 2, 3, 4, 5, 6, 7\}$

$$\begin{aligned}\text{No. of elements of the power set} &= 2^n \\ &= 2^7 \\ &= 128\end{aligned}$$

(iv) $\{0, 1, 2, 3, 4, 5, 6, 7\}$

$$\begin{aligned}\text{No. of elements of the power set} &= 2^n \\ &= 2^8 \\ &= 256\end{aligned}$$

(v) $\{a, \{b, c\}\}$

$$\begin{aligned}\text{No. of elements of the power set} &= 2^n \\ &= 2^2 \\ &= 4\end{aligned}$$

(vi) $\{\{a, b\}, \{b, c\}, \{d, e\}\}$

$$\begin{aligned}\text{No. of elements of the power set} &= 2^n \\ &= 2^3 \\ &= 8\end{aligned}$$

7. Write down the power set of each of the following sets:**(i) $\{9, 11\}$**

Let

$$\begin{aligned}A &= \{9, 11\} \\ P(A) &= \{\emptyset, \{9\}, \{11\}, \{9, 11\}\}\end{aligned}$$

(ii) $\{+, -, \times, \div\}$

Let

$$\begin{aligned}A &= \{+, -, \times, \div\} \\ P(A) &= \{\emptyset, \{+\}, \{-\}, \{\times\}, \{\div\}, \\ &\quad \{+, -\}, \{+, \times\}, \{+, \div\}, \\ &\quad \{-, \times\}, \{-, \div\}, \{\times, \div\}, \\ &\quad \{+, -, \times\}, \{+, -, \div\}, \{-, \times, \div\}, \\ &\quad \{+, \times, \div\}, \{+, -, \times, \div\}\}\end{aligned}$$

(iii) $\{\emptyset\}$

Let

$$\begin{aligned}A &= \{\emptyset\} \\ P(A) &= \{\emptyset, \{\emptyset\}\}\end{aligned}$$

(iv) $\{a, \{b, c\}\}$

Let

$$\begin{aligned}A &= \{a, \{b, c\}\} \\ P(A) &= \{\emptyset, \{a\}, \{\{b, c\}\}, \{a, \{b, c\}\}\}\end{aligned}$$